Academic year 2025-2026 (Odd Semester)

Date	**Dec 2025**	**Maximum Marks**	**50 + 10**
Course Code	**HS271TA**	**Duration**	**120 Min**
Sem	**VII**	**CIE II**	
INDIAN KNOWLEDGE SYSTEM			

PART – A (10 Marks)

Sl.No		M	CO	BT
1	**Which Doṣa is primarily responsible for movement in the body?** a) Pitta b) Kapha c) Vāta d) Tejas	1	3	L1
2	**The treatise that governs Vāstu Śāstra principles is:** a) Nāṭya Śāstra b) Mānasāra c) Sushruta Samhita d) Rigveda	1	3	L1
3	**The basic unit of Indian town planning in Vāstu is:** a) Maṇḍala b) Karmic grid c) Pada d) Prāṇa grid	1	3	L1
4	**The Eight Limbs of Vāstu are known as:** a) Aṣṭāṅga Vāstu b) Saddarśana c) Pañcha Mahābhūta d) Mantra Śāstra	1	2	L1
5	**The temple style with curvilinear Shikhara is:** a) Nagara b) Dravidian c) Vesara d) Ajanta	1	2	L1
6	**The ratio of circumference to diameter is known as:** a) e b) π c) φ d) α	1	3	L1
7	**Āryabhaṭa proposed that Earth:** a) Is stationary b) Rotates on its axis c) Orbits the Sun annually d) Is flat	1	2	L1
8	**Which metal was used first in ancient India?** a) Iron b) Copper c) Steel d) Aluminium	1	3	L1
9	**Which Yantra is used for measuring time?** a) Gola-yantra b) Chakra-yantra c) Water clock (Ghati Yantra) d) Shanku-yantra	1	3	L1
10	**The Chandasutras are fundamentally related to:** a) Geometry b) Meter of poetry c) Planetary motion d) Aromatic chemicals	1	2	L1

TEST

Sl. No.	Part- B Questions	M	CO	BT
1 a	Derive the nth term of an arithmetic progression	03	3	L1
b	Mention any two Panchanga elements.	02		
c	What is Wootz steel	02		
d	Explain Piṅgala's contribution to combinatorics	03		
2 a	Describe the role of geometry in temple construction.	03	3	L2
b	Describe the art of making natural perfumes.	03		
c	What is Piṅgala's binary system?	02		
d	What is a Tithi?	02		

Academic year 2025-2026 (Odd Semester)

3 a	Explain the features of Nagara and Dravidian temple architecture with examples.	03	2	L1
b	Describe major water management practices for sustainable agriculture	03		
c	Explain the BMI–Consciousness Complex with a suitable example	02		
d	Describe the Triguṇa System and its influence on human behaviour	02		
4 a	Compare Nagara and Dravidian temple architecture with examples.	03	2	L2
b	Describe the Triguṇa system and its psychological significance	02		
c	State components of the BMI-C model.	02		
d	Discuss the educational significance of the 64 Kalās.	03		
5 a	Mention any two of the 64 Kalās	02	3	L1
b	Differentiate between Tamas, Sattva and Rajas	03		
c	What is meant by Aṣṭāṅga Vāstu? Give two examples	03		
d	State two advantages of sprinkler irrigation.	02		

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	24	36	-	40	20	—	—	—	—

Course Outcomes:

CO1 Explain the nature, scope, and historical development of Indian Knowledge Systems and differentiate traditional knowledge from modern scientific paradigms

CO2 Analyze the foundational concepts of Indian contributions to linguistics, mathematics, and science, including works of Pāṇini, Piṅgala and others

CO3 Evaluate the applications of traditional Indian technologies in areas such as astronomy, metallurgy, architecture, and medicine.

CO4 Assess the relevance of Indian Knowledge Systems in the context of sustainable development goals and propose methods to safeguard and integrate IKS in contemporary society.

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15th December 2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : 7th	**UG**	**Duration : 30 + 90 Min**
Course Title: Product Design and Development		**Course Code : IM372IA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is concept generation in product design?	01	L_1	CO3
2.	Name one traditional method used for concept generation.	01	L_1	CO3
3.	Name the two main methods of concept selection.	01	L_1	CO3
4.	Name the types of product specifications.	01	L_2	CO4
5.	What is the role of AI-assisted benchmarking in concept testing?	01	L_2	CO3
6.	What is the first step in concept testing? Selecting the target customer segment.	01	L_1	CO4
7.	Why is variety and supply chain consideration important in product architecture?	01	L_2	CO3
8.	What is product architecture?	01	L_2	CO4
9.	What are implications of product architecture?	01	L_3	CO3
10.	What is Generative AI-based concept ideation?	01	L_2	CO4
	Part – B			
1a.	You are designing an autonomous delivery drone. Establish three critical product specifications that represent a conflict among technical feasibility, user experience (UX), and legal constraints. Propose a single, weighted trade-off solution for one of the conflicts.	05	L4	CO1
b.	Critically evaluate the balance required between the need for measurable, quantitative specifications (metrics) and the consideration of subjective user experience (UX) quality when defining the product specifications for a new predictive maintenance industrial IoT sensor.	05	L_4	CO1
2a.	What are some different ways you could communicate a concept for a new user interface for an automotive audio system? What are the strengths and weaknesses of each approach?	06	L_3	CO4
b.	Roughly estimate N for the following product. List your assumptions. • A sleeping pillow for air travelers. Assume that there are 2000 jumbo jets in service and that most sleeping pillows will be used on jumbo jets.	04	L_4	CO3
3	What are the items to be included in the survey format to gather data from potential customers of a new product conceived by the design team? Why do you think respondents typically overestimate the likelihood that they will purchase a product? What advantages Survey has over the customer interviews for Concept Testing?	10	L_4	CO3
4a.	How can a design team use concept generation to develop a sustainable packaging solution for e-commerce?	05	L_4	CO4
b.	How does generative AI support concept ideation in the automotive industry?	05	L_3	CO3

5a.	Evaluate the trade-off between modularity and integration in a complex electromechanical product (e.g., a smart vacuum cleaner). Under what specific market condition (e.g., low volume vs. high variety) would the integral architecture still be the superior choice?	06	L_4	CO4
b.	A design team needs to quickly select the best concept among five options. Justify the choice of using Concept Screening (vs. Scoring) first, not merely for resource saving, but as a critical tool for achieving team consensus and aligning expectations.	04	L_4	CO3

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Date : 15/12/2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VII	**UG**	**Duration : 30 + 90 Min**
Course Title: Total Quality Management		**Course Code : IM373TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Name any two QC tools commonly used in the seven-step problem-solving approach.	2	1	CO2
2.	Why is the catch-ball process important in Hoshin Management?	2	1	CO3
3.	Differentiate between proactive improvement and preventive action.	2	2	CO2
4.	A team finds recurring defects in a product. Which 8D steps should they focus on immediately to contain the issue and begin analysis?	2	3	CO3
5.	A manager wants to identify relationships among complex causes of a problem. Which Management and Planning Tool should they use?	2	3	CO4
	Part – B			
1	"The case study, discussed by Shoji Shiba, by the Broken Pellicle QI Team at Analog Devices illustrates the use of several TQM standards: the 7 steps, the most commonly used three of the 7 QC tools, and a standard presentation format called a quality improvement story or QI story". Provide a summary of learning from this case study.	10	4	CO2
2	How do you deploy semantics to distinguish between two kinds of language: affective language and the language of reports? Elucidate.	10	3	CO2
3	List and explain the seven principles for activating teamwork in an organisation.	10	2	CO3
4	Illustrate and evaluate the Proactive, Reactive, and Control Phases in Hoshin Management.	10	2	CO3
5	Evaluate the three categories of learning that are part of TQM learning system.	10	2	CO3

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Date : 16th Dec 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : 7th	**UG**	**Duration : 30 + 90 Min**
Course Title: Predictive Analytics		**Course Code : IM374TFA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What does accuracy measure in a classification model?	01	L_1	CO1
2.	What does precision measure in a classification model?	01	L_1	CO1
3.	In linear regression, what does the slope coefficient represent?	01	L_2	CO1
4.	Which metric is crucial in quality inspection models where missing defects is costly?	01	L_2	CO1
5.	Predicting employee attrition in HR analytics is an example of which of the six tasks od predictive analtyics?	01	L_2	CO1
6.	Which distance metric is commonly used in k-means clustering?	01	L_1	CO2
7.	What are the two main types of hierarchical clustering?	01	L_1	CO1
8.	Which clustering method does not require specifying the number of clusters in advance?	01	L_1	CO1
9.	What does “confidence” measure in association rules?	01	L_2	CO2
10.	What is the main limitation of Apriori algorithms?	01	L_2	CO2
	Part – B			
1 a.	State the primary objective function that the k-means algorithm attempts to minimize during the clustering process.	04	L_3	CO3
b.	Define lift as applied to association rule mining. In the context of market basket analysis, an analyst finds the rule Diapers→Beer has a high support and confidence, but a lift value close to 1.05. Explain what this lift value specifically indicates about the association between Diapers and Beer purchases.	06	L_3	CO3
2.	Write down the Mathematical form of the Sigmoid function and its graphical representation in the context of a Logistics Regression. Logistic Regression – Loan Approval Dataset: Income | Approved 30 | 0 50 | 1 70 | 1 Goal: Predict approval probability for Income = 60.	10	L_4	CO1
3.	What is the Significance of K in K-Means Algorithm? K-means with three clusters in 2D Data points: (X={(0,0),(0,3),(3,0),(3,3),(6,6)}) Initial centroids: (μ_1=(0,0), μ_2=(3,3), μ_3=(6,6)) Task: One assignment step; list clusters.	10	L_3	CO1

<table>
<tr><td>4 a.</td><td>What is the purpose of constructing a dendrogram to represent final clusters in Hierarchical Clustering?
Hierarchical clustering (single linkage) in 1D
Data points: (x={1,2,4,10})
Distance metric: Euclidean distance
Linkage: Single
Task: Build the dendrogram order and report clusters at threshold (t=3).</td><td>[illegible]</td><td>[illegible]</td><td>[illegible]</td></tr>
<tr><td>5.</td><td>Consider a small grocery store transaction database. Each transaction includes items [illegible] bought together. Let’s analyze the following dataset:
<table>
<tr><td>Transaction ID</td><td>Items Bought</td></tr>
<tr><td>1</td><td>Milk, Bread, Butter</td></tr>
<tr><td>2</td><td>Milk, Bread</td></tr>
<tr><td>3</td><td>Bread, Butter</td></tr>
<tr><td>4</td><td>Milk, Butter</td></tr>
<tr><td>5</td><td>Bread, Butter, Jam</td></tr>
</table>
i. Calculate Support for all the items in the itemset above.
ii. Generate Frequent Itemset
iii. Generate Association Rules and Calculate the confidence of these Association Rules.
iv. Calculate Lift.</td><td>[illegible]</td><td>[illegible]</td><td>[illegible]</td></tr>
</table>

[The right-hand M/BT/CO columns of Q4 and Q5 and anything after Q5 (iv) are cut off in the scanned copy.]

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Academic year 2025-2026 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**16.12.2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM375TGM**	**Duration**	**120 Mins**
Sem	**VII Semester**	**CIE – II**	
PROJECT MANAGEMENT (Institutional Elective)			

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	Why is a project life cycle important?	02	L1	CO1
2	What is the primary output of the Plan Quality Management process?	02	L2	CO1
3	Mention the key function of the Change Control Board (CCB).	02	L2	CO2
4	List two organizational factors that influence project management.	02	L1	CO1
5	What is the difference between corrective action and preventive action?	02	L2	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	During execution of a highway construction project, the project faces delay due to land acquisition issues. Explain how stakeholder management and governance frameworks help resolve such critical external dependencies.	10	L3	CO3
2.	Explain how different organizational structures (Functional, Matrix, Projectized) influence the decision-making speed and resource allocation of a project. Apply the concepts to a real-world example.	10	L2	CO2
3.	A consumer electronics company wants to launch a new Smart Home Security Device (AI-based door camera + alarm system). As the Project Manager, prepare a detailed Project Charter for this new product development project. Include all key components of a formal project charter and explain how each element is applied to this situation.	20	L4	CO4
4.	A company is developing an online Fee Payment Portal for a university. As the Project Manager, explain how you will apply the three processes of Project Quality Management — Plan Quality Management, Perform Quality Assurance, and Control Quality — to ensure high-quality software delivery. Provide suitable software-related examples for each stage.	10	L4	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	06	04	–	–	–	04	06	–	–	–	–
		Test		10	10	30	–		10	10	30	–	–

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Date : 16 Dec 2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VII	**UG**	**Duration : 30 + 90 Min**
Course Title: Principles & Practices of Cyber Law		**Course Code : HS375TGT**

Note: Answer all questions

Q.No.	PART- A	M	BT	CO
1	Define Digital Signature.	2	2	CO4
2	Shreya signed an online contract for a product, but later she wanted to dispute the contract terms since the contract was made electronically. Can the contract be challenged under the IT Act, 2000?	2	3	CO4
3	Give an example of technological advancement that led to the use of Electronic Signatures.	2	2	CO3
4	A suspect denies being present at a crime scene, but investigators suspect he used Google Maps that night. What mobile evidence can help verify his location?	2	3	CO3
5	What is the role of Interpol in cybercrime investigation?	2	2	CO4
	PART -B			
1	Explain the scope and importance of Digital Forensics in investigating cybercrimes.	10	02	CO3
2	A company discovers that sensitive financial files have been leaked. The IT team suspects that an employee copied the files to a USB drive before resigning. Investigators seize the employee's workstation and USB drives for analysis. a. Which type of forensic analysis should be conducted on the workstation? b. How can investigators determine if files were copied to the USB drive? c. Which tools can be used to acquire and examine digital evidence?	10	04	CO3
3	Discuss the importance of FIR and Charge Sheet in cybercrime prosecution .Explain the procedure for filing an FIR in cybercrime cases.	10	03	CO3
4	Discuss the relevance of signature in the electronic era, highlighting the difference between handwritten and digital signatures, and the technological advancements that have influenced their development.	10	03	CO2
5	A customer, Priya, purchases a laptop from an e-commerce website "TechBuy.in." She makes an online payment through a digital wallet, and the website issues an electronic invoice. However, after delivery, Priya discovers that the laptop is defective and that the return request she submitted through the website is ignored. When she contacts customer service, she receives no response. Priya files a complaint claiming that the e-commerce company failed in its legal duty under the *Information Technology Act, 2000* and violated her consumer rights. Based on this case, answer the following: (i) Explain how the *Information Technology Act, 2000* provides a legal framework for E-Commerce And Electronic Transactions. (ii) Discuss the key issues and challenges in regulating e-commerce in India as highlighted by this case. (iii) Suggest legal and practical measures that can protect consumers like Priya in the digital marketplace.	10	04	CO3

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Date : 16-12-2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VII	**UG**	**Duration : 30 + 90 Min**
Course Title: Sustainability Engineering & Circular Economy (Elective)		**Course Code : IM374TFD**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	PART – A			
1	What is a conceptual standard framework?	2	2	3
2	What is sustainable supply chain management?	2	2	3
3	List two impacts of climate change on engineering.	2	1	3
4	What is the 3R model in the context of circular economy?	2	2	4
5	What is meant by eco-design?	2	2	4
	PART – B			
1.	What are the key steps involved in formulating sustainability standards? How do international and national bodies collaborate to ensure these standards are credible, applicable, and effective across industries? Illustrate your answer with examples of standard-setting frameworks or organizations.	10	2	3
2.	Why is corporate sustainability becoming a strategic priority for modern enterprises? How does integrating environmental, social, and governance (ESG) principles enhance long-term business performance and stakeholder trust? Discuss with relevant examples and frameworks.	10	3	3
3.	How do global mega-trends such as urbanization, climate change, digitalization, and demographic shifts shape the future of sustainable engineering? In what ways must engineers adapt their design, operations, and decision-making to align with these evolving trends? Discuss with relevant examples and implications.	10	4	3
4.	How do smart technologies such as IoT, AI, and blockchain contribute to the implementation of circular economy principles? In what ways can these technologies support resource optimization, waste reduction, and closed-loop systems in engineering and industrial contexts?	10	4	4
5.	What is a digital factory, and how does it transform traditional manufacturing systems? In what ways does the integration of digital technologies in factory operations contribute to sustainability?	10	3	4

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 16th Dec 2025	**CIE-II**	**Max. Marks : 10 + 50**
Semester : VII	**UG**	**Duration : 30 + 90 Min**
Course Title: Artificial Intelligence in Industrial and Systems Engineering		**Course Code : IM374TFB**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Traditional optimization techniques often rely on fixed mathematical models, whereas AI-based optimization uses _____ to learn pattern	1	3	01
2	AI-driven demand forecasting reduces forecasting error by using large-scale _____ data	1	2	01
3	Reinforcement learning is often used in production scheduling to maximize cumulative _____.	1	2	02
4	AI-based inventory management helps determine safety stock using real-time ____ data.	1	3	02
5	In logistics optimization, route planning can be enhanced through AI using ___ algorithms.	1	2	02
6	In automotive quality control, ____ vision is widely used for surface defect detection.	1	3	01
7	Aerospace condition monitoring makes use of ____ frequency analysis to detect engine issues.	1	2	01
8	Time-series forecasting in predictive maintenance frequently uses _____ models.	1	2	02
9	Image-based defect detection systems require accurate _____ to mark true defects.	1	2	03
10	Feature extraction in anomaly detection transforms raw signals into meaningful _____.	1	2	02
	Part – B			
1	Explain AI-powered approaches for demand forecasting and inventory management.	10	3	01
2	Discuss AI-based production scheduling and resource allocation techniques with examples.	10	3	02
3	Describe the limitations of traditional optimization techniques in supply chain management.	10	2	03
4	Explain the importance of sensor fusion in predictive maintenance.	10	3	03
5	Explain end-to-end predictive maintenance using ML: data, models, diagnostics, forecasting, and decision-making.	10	2	03

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Department of Electrical & Electronics Engineering

Date: 16/12/2025	**Test - 2**	**Max. Marks:50+10**
Semester:7	**UG**	**Duration: 1 H:50 M**
Course Title: e-Mobility		**Course Code:EE375TGH**

SL No.	PART-A QUIZ QUESTIONS	M	BT	CO
1	List the essential functions of BMS.	1	L1	CO2
2	A 100 Ah battery delivered 20 Ah. Calculate SOC.	1	L2	CO2
3	List any two electrochemical reactions involved in a Li-ion battery	1	L2	CO1
4	State any two requirements of energy storage in hybrid electric vehicles.	1	L1	CO1
5	What is the key advantage of supercapacitors over batteries?	1	L1	CO2
6	What is hybridization of energy storage systems? Give one example.	1	L1	CO3
7	Differentiate between SOC (State of Charge) and SOH (State of Health)	2	L1	CO3
8	Differentiate between battery-based and fuel-cell-based energy storage systems.	2	L2	CO2

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PART-B TEST QUESTION

		M	BT	CO
1. a	Explain the redox reaction in a battery? Explain the basic structure of a Battery.	6	*[illegible]*	*[illegible]*
b	What is the voltage range for the cell when used in a HEV application with a DOD of 25% to 75% and a load ranging from no-load to a full load of 6C? What is the battery pack voltage if there are 192 cells arranged with 96 cells in series and two strings in parallel?	4	*[illegible]*	*[illegible]*
2. a	Explain the following regarding the charging of Lithium-ion batteries (a) Influence of Charging voltage on charging process. (b) Influence of charge current on charging process.	4	*[illegible]*	*[illegible]*
b	Explain the different types of Battery Management System (BMS) topologies Centralized, Modular, and Distributed .along with their advantages and disadvantages.	6	*[illegible]*	*[illegible]*
3. a	With neat schematic explain the concept of Active balancing and passive balancing of batteries.	6	*[illegible]*	*[illegible]*
b	A 24 kWh battery pack can be fast charged from 0% to 80% SOC in 30 min. Determine the approximate charge current and power in order to achieve this charge time.	4	*[illegible]*	*[illegible]*
4. a	Draw and explain the block diagram of a complete SRM drive system including the motor, converter, controller, and feedback loops. Discuss how each block contributes to stable and efficient operation.	10	*[illegible]*	*[illegible]*
5. a	. With neat block diagram, explain the working of a Fuel Cell Energy Storage System for EV propulsion.	6	*[illegible]*	*[illegible]*
b	List any four performance parameters used to evaluate energy storage systems in EVs.	*[illegible]*	*[illegible]*	*[illegible]*

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DEPARTMENT OF MECHANICAL ENGINEERING AY 2025-26 (ODD SEM)				
Date	**16/12/2025 Friday 1.20 PM to 3.20 PM**	**CIE 2**	**Max. Marks**	**10+50**
Semester	**VII**	**UG**	**Duration**	**2 Hrs.**
Course	**INDUSTRY 5.0 (Institutional Elective)**		**Course Code**	**ME375TGP**

#	QUIZ	M	BT	CO
1	Which statement best describes a digital twin?	1	1	1
2	Mention a key feature that distinguishes a digital twin from a standard simulation.	1	1	1
3	Highlight a security challenge associated with digital twins.	1	2	2
4	The use of AI-driven supply chain optimization leads to which benefit?	1	2	2
5	What is a major ethical concern regarding the increased use of AI and automation in manufacturing?	1	2	2
6	What is edge computing?	1	1	1
7	How does a circular economy primarily address greenhouse gas emissions?	1	2	2
8	What is the term for a systematic approach that assesses a product's environmental impact throughout its entire lifespan, from raw material extraction to disposal?	1	1	1
9	What are the Sustainable Development Goals (SDGs)?	1	2	2
10	What is the primary focus of the Global Reporting Initiative (GRI)?	1	2	2

#	TEST	M	BT	CO
1	Explain the working of digital twins. Also discuss the Key applications and benefits across industries	10	2	3
2	What do you mean by Artificial Intelligence (AI) & Machine Learning (ML)? Explain the application of AI and ML in (i) Predictive maintenance (ii) Supply chain optimization (iii) Product development and innovation (iv) Human-robot collaboration	10	4	2
3	What are Cyber-Physical Systems? Discuss the core components of CPS and the applications of CPS	10	2	*[illegible]*
4	Briefly discuss the principles of (i) Sustainable and Resilient manufacturing (ii) Circular economy (iii) Life cycle assessment	10	3	*[illegible]*
5	Explain the regulatory frameworks and standards for sustainable practices with examples.	10	4	*[illegible]*

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Date: 16-12-25	**CIE-II**	**Max. Marks: 10+50**
Semester: VII	**UG**	**Duration: 2 Hrs**
Course Title: Statistical Methods for Engineers		**Course Code: MA375TGO**

Department of Mathematics

Sl. No.	QUIZ	M	BT	CO
1	You are performing an ANOVA Analysis, and the total sum of squares is 36 and the treatment sum of squares is 16, what would the error sum of squares be?	1	2	1
2	In ANOVA analysis, the degrees of freedom associated with the numerator of the F-value are calculated using which formula?	1	1	1
3	The one-way ANOVA Analysis below has 10 treatment groups with the total degrees of freedom of 19. Variation Source | Sum of Squares (SS) | Degrees of freedom (DF) | Mean Squares (MS) | F - Value Treatment(Between) | | | | | Error (Within) | 55 | | | | Total | 100 | 19 | | | Calculate the Treatment Mean Square for this ANOVA Table.	2	2	4
4	In a 2^3 factorial experiment with three factors A, B and C, each at two levels (0 and 1), write the treatment combinations for: (i) A and C at the first level and B at the second level. (ii) A and C at the second level and B at the first level.	2	1	3
5	Classify the data point (2.5, 5.5), given the linear score functions for two populations as $\hat{S}_a^L(x) = -441.6006 + 156.3833\,x_1 + 63.3446\,x_2$ $\hat{S}_b^L(x) = -292.5690 + 127.5972\,x_1 + 51.3522x_2$.	2	1	1
6	Given the covariance matrix for a data with 3 variables X_1, X_2, X_3 as $\begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and the first principal component as $\begin{bmatrix} 0.383 \\ -0.924 \\ 0 \end{bmatrix}$, then the correlation of X_1 with Y_1 and correlation of X_2 with Y_1 are ____ and ____ respectively. *(handwritten on the original: λ's are 5.83, 2, 0.17)*	2	1	2

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<table>
<tr><th>Sl. No.</th><th>TEST</th><th>M</th><th>BT</th><th>CO</th></tr>
<tr><td>1</td><td>A local bank has three branch offices. The bank has a liberal sick leave policy, and a vice-president was concerned with employees taking advantage of this policy. She thought that the tendency to take advantage depended on the branch at which the employee worked. To see if there were differences in the time employees took for sick leave, she asked each branch manager to sample employees randomly and record the number of days of sick leave taken during 1990. Ten employees were chosen, and the data are listed below. Does this data indicate a difference in branches? Use a level of significance of 0.05. [Given $F_\alpha(df_{k-1}, df_{N-k}) = 4.74$]
<table>
<tr><td>Branch 1</td><td>15</td><td>20</td><td>19</td><td>14</td></tr>
<tr><td>Branch 2</td><td>11</td><td>15</td><td>11</td><td></td></tr>
</table>
[illegible] (the rest of the data table is cut off in the scan)</td><td>10</td><td>2</td><td>[illegible]</td></tr>
<tr><td>2</td><td>[illegible] researcher wants to test four diets A, B, C, D on growth rate in [illegible] are divided into 3 groups according to their weights. Heaviest 4, [illegible] put in Block I, Block II and Block III respectively. Within each block, [illegible] is given at random to the animals. After 15 days, increase in weight is [illegible] given in the following table:
<table>
<tr><th>Blocks</th><th colspan="4">Treatments/Diets</th></tr>
<tr><th></th><th>A</th><th>B</th><th>C</th><th>D</th></tr>
<tr><td>I</td><td>12</td><td>8</td><td>6</td><td>5</td></tr>
<tr><td>II</td><td>15</td><td>12</td><td>9</td><td>6</td></tr>
<tr><td>III</td><td>14</td><td>10</td><td>8</td><td>5</td></tr>
</table>
Perform a two-way ANOVA to test whether the data indicate any significant [illegible] between the four diets due to different blocks. Use a level of significance [illegible] [Given: $F_{[illegible]}(2,6) = 10.925$ & $F_{[illegible]}(3,6) = 9.780$]</td><td>[illegible]</td><td>[illegible]</td><td>[illegible]</td></tr>
<tr><td>3</td><td>[illegible] experiment was planned to study the effect of Sulphate, Potash and Super[illegible] on the yield of potatoes. All the combinations of 2 [illegible] levels of Super[illegible] [0 cent (p_0) and 5 cent (p_1) [illegible] acre] and two levels of Sulphate and Potash [illegible] (k_0) and 5 cent (k_1) [illegible] acre] were studied in a randomized block design with [illegible] replications each. The yields (in appropriate units) obtained are given in the table below.
[Table "Yields (lbs per plot)" by Blocks and Treatment combinations: illegible]
[illegible] the data of a 2^2 factorial experiment conducted using a RBD and give your [illegible]. Use a level of significance of [illegible] Given: F[illegible] = 5.12]</td><td>[illegible]</td><td>[illegible]</td><td>[illegible]</td></tr>
<tr><td>4</td><td>[illegible] Linear Discriminant [illegible] the following two dimensional dataset: [illegible] and hence classify the data point [illegible]</td><td>[illegible]</td><td>[illegible]</td><td>[illegible]</td></tr>
<tr><td>5</td><td>[illegible] Principal Component Analysis of the data and hence compute the percentage [illegible] variance contained in each principal component [illegible]</td><td>[illegible]</td><td>[illegible]</td><td>[illegible]</td></tr>
</table>

[Test Q2–Q5 are on a scanned page that is too blurred to read in full.]

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Department of Civil Engineering

Date: 16-12-2025	CIE- II	Max. Marks: 60
Semester: VII	UG	Duration: 2 Hrs
Course Title: Freight Transportation Systems and Logistics		Course Code: CV374TGH

Quiz

S No	Questions	M	BT	CO
1.	Explain the term freight capacity issues with an example.	2	2	2
2.	How does freight productivity influence overall logistics performance?	2	2	2
3.	Briefly describe the role of scheduling in freight transportation.	2	2	*[illegible]*
4.	List any two current trends in logistics and explain them.	2	2	*[illegible]*
5.	What is the aim of logistics in a supply chain context?	2	2	3

S No	Questions	M	BT	CO
1.	Explain the factors affecting freight productivity and performance. Illustrate how organizations measure these factors.	10	3	*[illegible]*
2.	Describe the process of route planning, scheduling, collection, storage, and distribution center operations in logistics networks.	10	3	2
3.	Explain strategic decisions in logistics and discuss the steps involved in designing an effective logistics strategy.	10	3	3
4.	Compare various competing modes of transport for specific commodities. How does mode selection affect cost and service levels?	10	3	2
5.	Evaluate the importance of distribution centers in improving freight performance. Provide examples of their strategic roles.	10	3	3

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Department of Electronics and Telecommunication Engineering

Date: 16/12/25	**CIE - 2**	**Max. Marks: 60**
Semester: 7	**UG**	**Duration: 2 Hrs**
Course Title: Space Technology and Applications		**Course Code: ET375TGL**

Q.No	Quiz-2	M	BT	CO
1	List the classification of satellites based on service.	2	L2	CO2
2	Mention the difference between linear bent pipe transponder and baseband processing transponder.	2	L2	CO2
3	What is the main function of command systems in satellites?	2	L2	CO2
4	What is multiple access? Mention its advantages.	2	L2	CO3
5	What is a transponder?	2	L2	CO3
	Test-2			
1	Illustrate the altitude and orbit control subsystem present in the satellite with a neat diagram.	10	L3	CO2
2	With a neat diagram, explain the linear bent pipe transponder.	10	L2	CO2
3a	Explain the functions of tracking systems in the satellite. List and explain the different types of tracking methods.	6	L2	CO2
3b	What are the main functions of telemetry and monitoring systems?. List the various data collected to check the status of the satellite.	4	L2	CO2
4	Explain the different types of multiple access techniques used for satellite communication systems.	10	L2	CO3
5	With a neat block diagram, explain the DBS receiver system.	10	L2	CO3

BT- Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	36	24	-	-	50	10	-	-	-

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Department of
Aerospace Engineering

Date	14th December 2025	Maximum Marks	60
Course Code	AS375TGB	Duration	120 Min
Course Title	AIRCRAFT SYSTEMS	Semester	VII
ACY	2025-26	Quiz &Test-II	

PART -A

Sl No	Questions	M	BT	CO
1	What is the primary purpose of an APU in an aircraft during ground operations?	2	2	3
2	Differentiate between electric and pneumatic starting systems.	2	2	2
3	Why is it important to achieve a minimum rpm before introducing fuel during start?	2	3	1
4	Name any two major components of an aircraft fuel system and state their functions.	2	3	3
5	What triggers the deployment of a RAT in modern aircraft?	2	2	4

PART-B

Sl.No.	Questions	M	BT	CO
1	Describe the working principle of Pressure Feed Fuel Systems with required sketch.	10	L2	2
2	Explain Briefly about the Magneto – Ignition System & FADEC system	10	L1	1
3	With a neat sketch explain wet sump lubrication system	10	L2	3
4	Explain Pressure measurements and indicating systems with a neat sketch	10	L3	3
5	Explain Mach Warning system and altitude alerting System	10	L3	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	12	12	24	12	10	38	24	0	0	

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Department of Civil Engineering

Date	16.12.2025	CIE – II (Test + Quiz)	Maximum Marks	50+10
Semester	VII	UG (Institutional Elective)	Duration	90+20 Min
Solid Waste Management and Statutory Rules			Course Code : CV375TGG	

Note: Answer all questions. Assume missing data if any suitably.

Q.No.	**PART A** Questions	M	BT	CO
1.	State the characteristics of Hazardous waste.	2	2	*[illegible]*
2.	Classify the methods of landfilling.	2	3	1
3.	Compare aerobic and anaerobic composting.	2	3	4
4.	Define Leachate and Hazardous waste.	2	1	3
5.	List any 4 factors influencing anaerobic composting process.	2	2	4

Q No.	**PART B** Questions	M	BT	CO
1.a	List and explain classification of Hazardous Waste with examples.	5	2	3
1.b	With the help of neat sketch enumerate the characteristics of onsite Hazardous Waste storage bin.	5	*[illegible]*	*[illegible]*
2	Enumerate salient features of Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016 rules	10	2	1
3	With the help of screening model illustrate identification of HW. Explain terminologies.	10	3	3
4	With the help of neat sketch explain control of gas movement by permeable methods.	10	3	4
5a	Determine the amount of oxygen required to compost 1000kg of organic solid waste aerobically. Assume that initial composition of the organic material to be decomposed is given by $[C_6H_7O_2(OH)_3]_5$ and the final composition is expected to $[C_6H_7O_2(OH)_3]_2$ and that 400Kg of material remains after composting process. Prove the same with proper check.	5	3	3
5b	List and briefly explain the factors to be considered in evaluating potential landfill sites.	5	*[illegible]*	*[illegible]*

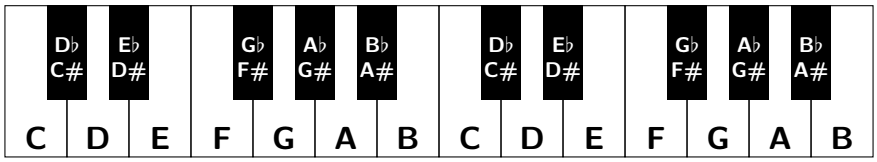
BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	12	5	27	16	02	29	29	-	-	-

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Date: 16-12-2025	CIE 2	Max. Marks: 10+50
Semester: VII	UG	Duration: 2 Hrs (1:20PM- 3:20PM)
Course Title: Mathematics of Music		Course code: MA375TGR

Department of Mathematics



S No	Quiz	M	BT	CO
1.	Construct the swara sequence for the 52nd Melakartha (Ramapriya).	2	1	2
2.	Write the continued fraction expansion of $log_2(3/2)$ up to three terms.	2	2	1
3.	Give the Fourier series form of a periodic function and explain what the coefficients represent physically.	2	2	2
4.	Define the spectral centroid and compute it for a harmonic spectrum with amplitudes $A = \{5, 3, 1\}$ at frequencies $\{f, 2f, 3f\}$.	2	3	3
5.	Define a pitch class and express pitch class addition formally in modulo 12.	2	2	3

S No	Test	M	BT	CO
1a	Prove that $log_2(3/2)$ is irrational using proof by contradiction. Explain in detail the musical consequence of it.	10	2	1
2a	Derive the formula for (or explanation on) the total number of Melakartha rāgas (72) starting from the combinatorial possibilities of swaras, and elaborate on the *Katapayadi sutra* therein.	10	3	2
3	Derive the mathematical explanation for difference tones using the nonlinear distortion equation: $y = ax + bx^2$. If two pure tones at frequencies f_1 and f_2 are input, show that the output contains the difference frequency $\lvert f_1 - f_2 \rvert$. When two notes at 600 Hz and 800 Hz are played simultaneously: (i) Calculate the first-order difference tone. (ii) Calculate the first-order summation tone. (iii) Identify the musical interval represented by 600 Hz and 800 Hz. Also, explain why combination tones are more prominent in organs and brass instruments.	10	3	4
4	For a square wave $f(t) = \begin{cases} k \; for \; 0 < t < \frac{T}{2} \\ -k \; for \; \frac{T}{2} < t < T \end{cases}$, derive its Fourier series representation showing that it contains only odd harmonics with amplitudes decreasing as $1/n$. Explain why the square wave sounds different from a sawtooth wave despite both being periodic.	*[illegible]*	*[illegible]*	*[illegible]*
5	Prove that the set of all transposition operations $\{T_0, T_1, T_2, \ldots, T_{11}\}$ forms a cyclic group under composition, isomorphic to $(\mathbb{Z}_{12}, +)$. Show that: (i) Closure: $T_m \circ T_n = T_{(m+n)} \; mod \; 12$ (ii) Associativity holds (iii) Identity element exists (iv) Inverse elements exist	*[illegible]*	*[illegible]*	*[illegible]*

Academic year 2025-2026(ODDSem)

DEPARTMENT OF

ELECTRONICS AND INSTRUMENTATION ENGINEERING

Date	**16.12.2025**	**Maximum Marks**	**10+50**
Course Code	**EI375TGK**	**Duration**	**120 Min**
Sem	**VII**	**CIE (Quiz &Test) - 2**	
Course Title: DISEASE AND DIAGNOSTICS AN ENGINEERING PERSPECTIVE			

Sl. No.	PART-A	M	BT	CO
1.1	What is spirometry?	1	L1	2
1.2	Define Tidal Volume (TV) and Residual Volume (RV).	2	L1	1
1.3	What is Functional Residual Capacity (FRC)? Mention its components.	2	L2	1
1.4	Which lung capacity cannot be measured using spirometry?	1	L1	1
1.5	Name two types of dialysis.	1	L1	2
1.6	Define kidney filtration.	1	L1	3
1.7	What is a dialyzer?	1	L1	3
1.8	What is creatinine?	1	L1	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	05	02	03	-	8	2	-	-	-	-

****

Sl. No.	PART- B	M	BT	CO
1.	Explain the functioning of the Respiratory System (RS).	10	L1	1
2.	Explain lung volumes and capacities with a neatly labeled diagram.	10	L2	2
3.	Describe any four common diseases of the Renal System.	10	L1	3
4.	Explain the structure and functioning of the Renal System.	10	L2	1
5.	Describe common ailments of the respiratory system, their causes, and symptoms.	10	L2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	20	10	20	-	20	30	-	-	-	-

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[Indian Knowledge System – a second CIE-II test page. The header and Part A of this paper are not in the scanned copy.]

TEST

Sl. No.	Part- B Questions	M	*[illegible]*
1 a	List the six classical schools of Indian philosophy.	03	
b	Explain any two threats to traditional knowledge.	02	
c	Mention one key difference between traditional Indian and Western systems of knowledge.	02	*[illegible]*
d	How does the Vedic Corpus form the foundation of Indian knowledge traditions?	03	
2 a	Draw a neat framework for classification of IKS Corpus	03	
b	Name the six Vedangas.	03	
c	State the two types of perception recognized in Nyaya philosophy	02	*[illegible]*
d	Differentiate between agricultural and ecological traditional knowledge.	02	
3 a	Mention any four Yogasutra of Patanjali	03	
b	Describe the characteristics and scope of Indian traditional knowledge in the modern context.	03	1
c	Mention one key feature that differentiates Nyaya from Samkhya philosophy.	02	
d	Name the six orthodox (*āstika*) schools of Indian philosophy	02	
4 a	Find out the difference from the Kaṭapayādi system to Bhūta-saṅkhyā for the word "Rāma".	03	
b	Name the three generic measures used in the ancient Indian system.	02	2
c	Laghu and Guru syllables in the given word: योगदर्शन (yōgadarśana)	02	
d	Develop a binary cycle of length four and identify all possible binary words of length fours.	03	
5 a	What is the standardized units for street widths in Indus-Saraswati civilization?	02	
b	Who the concept of zero discovered in India? What is its special significance?	03	
c	What is the numeral representation for the word 'manu, agni, elephant, Tithi, rudra, Indra, Pandava.	03	3
d	Briefly provide the organization of "Ashtadhyayi". How many rules are there in Ashtadhyayi ?	02	

[The heading of the fourth column and the column after it are cut off in the scan.]

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	10	5	—	15	10	—	—	—	—

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**27th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21HS71**	**Duration**	**120 Mins**
Sem	**VII Semester**	**CIE – II**	
CONSTITUTION OF INDIA AND PROFESSIONAL ETHICS			

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	Define Uniform Civil Code.	02	2	2
2	List any two powers of National Human Rights Commission (NHRC) .	02	2	2
3	What is “product liability”?	02	2	2
4	Differentiate between unfair Trade practice & Deficiency in Services.	02	2	2
5	List any two consumer Rights available under the Act	02	2	2

PART – B

Sl. No.	Questions	M	BT	CO
1.a	Examine the Constitutional position of Governor of a State.	05	2	1
1.b	Discuss Online shopping and product liability covering-Liability of a product retailer or delivery service and Liability related to online food products.	05	2	1
2.a	Explain the Impeachment proceedings of President of India.	05	2	2
2.b	Discuss the ground and procedure of removing the Judges of Supreme Court	05	3	2
3	List 11 Fundamental Duties enshrined in the Constitution of India.	10	2	1
4	The slogan line of “Jago Grahak Jago” is applicable today. Discuss the Historical perspective, Significance and Key features of Consumer Protection Act (both old and new).	10	3	2
5	Directive principle is fundamental in making Welfare state”. Discuss	10	4	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	Quiz	–	10	–	–	–	–	10	–	–	–	–
		Test	20	20	10	-	25	15	10		-	-	20

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**27th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM72**	**Duration**	**120 Mins**
Sem	**VII Semester**	**CIE – II**	
PRODUCT DESIGN & DEVELOPMENT			

Note:

1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.

PART – A

Sl. No.	Questions	M	BT	CO
1	What does Design for Cost (DfC) aim to achieve?	01	L_2	CO3
2	List any two main objectives of Design for Maintainability (DfM)?	01	L_2	CO3
3	Name one method used in Design for Usability (DfU).	01	L_1	CO3
4	Mention the purpose of Design for Sustainability (DfS)?	01	L_2	CO3
5	How does Design for Logistics (DfL) impact the supply chain management?	01	L_2	CO3
6	What is the focus of Design for Compliance (DfC)?	01	L_1	CO4
7	Mention Design for Testability (DfT) improve product quality?	01	L_2	CO4
8	Define “product backlog” in agile.	01	L_1	CO4
9	Which agile role is responsible for maximizing the value of the product?	01	L_1	CO4
10	Mention any one feature of the stage gate product development process.	01	L_2	CO4

PART – B

Sl. No.	Questions	M	BT	CO
1.	Explain the principles of Design for Assembly (DfA). How does DfA impact the efficiency and cost of product assembly? Use case studies or examples to support your answer.	10	L_3	CO3
2.	A company is developing a new eco-friendly packaging solution. Evaluate how Design for Sustainability (DfS) principles can be integrated into their product development process and the potential impact on their brand and market position.	10	L_4	CO4
3.	A consumer electronics company is launching a new smartphone and wants to ensure it is easy to assemble and test. Discuss how Design for Assembly (DfA) and Design for Testability (DfT) principles can be applied to achieve these goals.	10	L_4	CO4
4.	A global automotive company is developing an electric vehicle. Discuss how they can incorporate Design for Cost (DfC) and Design for Sustainability (DfS) principles to ensure the vehicle is affordable and environmentally friendly.	10	L_4	CO4
5.	Discuss the advantages and disadvantages of using the agile product development approach in a fast-paced technology environment. Provide examples to support your arguments.	10	L_3	CO3

Academic year 2024-2025 (ODD Sem)

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	Quiz	-	-	05	05	–	4	6	-	–	–	–
		Test			20	30	–	–	–	20	30		–

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Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**29th January 2025**	**Maximum Marks**	**50 + 10**
Course Code	**21IM74HD**	**Duration**	**90 + 20 Min**
Sem	**VII**	**CIE – II**	
DATABASE MANAGEMENT SYSTEM			

Note: Answer all the Questions.

Sl No	Questions	M	BT	CO
	Part A			
1.	Enumerate the three types of relationship that exists between entities in an ER diagram.	02	L1	CO2
2.	Mention the need for defining Database constraints in DBMS.	02	L4	CO1
3.	List the features of Foreign Key.	02	L2	CO3
4.	PROJECT is Unary operation. Justify	02	L4	CO3
5.	Consider the following schema of relations: *Student (ssn, name, address, major)* *Course (code, title)* *Registered (ssn,code)* Write an appropriate query for listing the codes of courses in which at least one student is registered (registered courses).	02	L3	CO4
	Part B			
1 a	Consider the following schema: Suppliers (sid : integer, sname : string, address :string) Parts (pid : integer, pname : string, color : string) Catalog (sid : integer,pid : integer, cost : real) i. Find the name of suppliers who supply some red parts ii. Find the sids of suppliers who supply every part	05	L3	CO4
b	Explain the integrity constraints: Not Null, Primary Key with an example each. Is the combination 'Not Null, Primary Key' a valid combination? Justify.	05	L4	CO3
2	Illustrate different types of Constraints in DBMS with examples.	10	L3	CO1
3	"Whenever update operations are applied on the relations, it is necessary that the integrity constraints are not violated". Evaluate the statement with relevant examples.	10	L4	CO3

Academic year 2024-2025 (Odd Sem)

<table>
<tr><td>4</td><td>Consider the following relation, BOOK.
BOOK<table><tr><th>Acc-No</th><th>Yr-Pub</th><th>Title</th></tr><tr><td>734216</td><td>1982</td><td>Supply Chain Management</td></tr><tr><td>237235</td><td>1995</td><td>Database design</td></tr><tr><td>631523</td><td>1992</td><td>Machine design</td></tr><tr><td>543211</td><td>1991</td><td>Quality Management</td></tr><tr><td>376112</td><td>1992</td><td>Personnel Management</td></tr></table>Write the syntax for applying the following queries on the relation, BOOK.
a) Select from the relation, BOOK all the books whose year of publication is 1992.
b) Select from the relation, BOOK all the books whose Acc-No is greater than or equal to 56782.
c) List all the Title and Acc-no of the BOOK relation.
List all the Title and Yr-Pub of the BOOK relation</td><td>10</td><td>L3</td><td>CO4</td></tr>
<tr><td>5</td><td>From the relations R and S , find the following:
(a) R ∪ S (b) R ∩ S (c) R – S (d) S – R
R<table><tr><th>Sl.No</th><th>DEPT</th></tr><tr><td>S1</td><td>PHYS1</td></tr><tr><td>S2</td><td>PSY</td></tr><tr><td>S3</td><td>CHEM</td></tr><tr><td>S4</td><td>JOUR</td></tr></table>S<table><tr><th>Sl.No</th><th>DEPT</th></tr><tr><td>S10</td><td>MATH1</td></tr><tr><td>S3</td><td>CHEM</td></tr><tr><td>S15</td><td>ENG</td></tr><tr><td>S16</td><td>MATH2</td></tr></table></td><td>10</td><td>L4</td><td>[illegible]</td></tr>
</table>

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	2	2	4	2	-	02	02	02	04	-	-
	Test		10	-	15	25	-	-	-	25	25	-	-

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**28.01.2025**	**Maximum Marks**	**10+50**
Course Code	**21IM74HA**	**Duration**	**30+90 Min**
Sem	**VII**	**CIE – II**	
FACILITIES PLANNING and DESIGN			

Note:

1. Answer all the Questions.

Q.No	Part – A Questions	M	CO	BT
1.1	Define a unit load as per Tanchoco et al.	(02)	L1	CO1
1.2	Small loads _____ the transportation requirements but can potentially _____ work-in-process inventory.	(02)	L2	CO1
1.3	List any two basic methods of moving the unit load according to Apple.	(02)	L1	CO1
1.4	What are the critical types of information needed to integrate a plant-wide material handling system	(02)	L1	CO1
1.5	Write in one sentence the JIT Impact on Facilities Design.	(02)	L1	CO1

Sl. No.	Part –B Questions	M	BT	CO
1	Enumerate and elaborate the steps in material handling system design.	10	L2	CO1
2	Given the following dimensions of a particular type of plastic reusable containers Inside dimensions 18” 11” 11” Outside dimensions 20” 12” 12” Each nested container 20” 12” 2” A trailer with inside dimensions of 240” 120” 120” is used to transport these containers. The containers are not palletized. Assume that no clearance is needed between containers or between containers and the walls of the trailer. The dimension of the storage opening is 24” 16” 14”. Determine the following: i. Container space utilization ii. Storage space efficiency iii. Container nesting ratio iv. Trailer space utilization if all containers are stacked vertically in only one orientation v. Trailer return ratio	10	L3	CO3
3	Explain the guidelines for estimating the material handling costs in practice.	10	L2	CO2
4	Summarize the features of fixed automation systems. Evaluate the merits, demerits and applications of such systems.	10	L3	CO3
5	List the manufacturing situations for which a flexible manufacturing system (FMS) might be appropriate. What are the various means of achieving flexibility in FMS? List and illustrate with sketches the alternative layouts for a FMS.	10	L1	CO1

Academic year 2024-2025 (ODD Sem)

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	10				8	2				
	Test		20	10	20		10	20	20			

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**29th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM75IM**	**Duration**	**30 + 90 Min**
Sem	**VII**	**CIE – II**	
PROJECT MANAGEMENT (Institutional Elective-II)			

Note:
1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	Mention the inputs required to create project charter process.	02	L1	CO1
2	What is work performance data?	02	L2	CO1
3	Give any two examples of prevention costs in cost of quality.	02	L3	CO2
4	Mention the use of affinity diagram in managing the quality.	02	L2	CO1
5	What is secondary risk? Give an example.	02	L2	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	Assume that you are developing a **Student Management Information System** (SAP) for your institution. Create a project charter for the above case. Also discuss what are the issues with the current SAP system.	15	L4	CO3
2.	You are leading a project to develop a new e-commerce platform. During the initial stages, your stakeholders are unclear about the scope and objectives. a) Identify the key document you need to create to formally authorize the project and outline its purpose. b) Explain the process you would follow to ensure all stakeholders are aligned with the project scope and objectives.	10	L3	CO4
3.	Compare and contrast qualitative and quantitative risk analysis techniques used in project risk management, providing examples of when each approach is most appropriate.	10	L3	CO2
4.	.Evaluate the potential consequences for a company that neglects to address quality costs in its operations, considering both short-term and long-term implications. Take a case example of any project of your choice.	10	L3	CO3
5.	List seven quality tools used to manage project.	05	L1	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars			CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks Distribution	Max Marks	Quiz	06	04	–	–	–	02	06	02	–	–	–
		Test	–	15	25	10	–	05	–	30	15	–	–

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**28th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM73G1**	**Duration**	**120 Mins**
Sem	**VII Semester**	**CIE – II**	
DATA MINING AND PREDICTIVE ANALYTICS (Elective)			

Note:
1. Answer all the Questions.
2. Avoid verbose answers and answer specifically and to the point.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is Association Rule Mining in data mining?	01	L_2	CO2
2	Which metric measures the strength of the association rule?	01	L_1	CO2
3	Define the term 'support' in the context of Association Rule Mining.	01	L_1	CO2
4	What does the acronym CART stand for in data mining?	01	L_1	CO2
5	What does the term 'confidence' refer to in Association Rule Mining?	01	L_2	CO2
6	What type of variable does a regression tree predict?	01	L_2	CO3
7	What does 'Gini index' measure in decision tree analysis?	01	L_2	CO3
8	In K-Means clustering, 'K' stand for_____	01	L_2	CO3
9	What metric is commonly used to measure distance in K-Means Clustering?	01	L_2	CO3
10	Give one limitation of the K-Means Clustering algorithm.	01	L_2	CO3

PART – B

Sl. No.	Questions	M	BT	CO
1.	Explain the Apriori algorithm used in association rule mining. Discuss its importance and provide a step-by-step example to illustrate its working.	10	L_3	CO2
2.	Given a dataset of transactions: - Transaction 1: {A, B, D} - Transaction 2: {B, C} - Transaction 3: {A, B, D, E} - Transaction 4: {A, C, D} - Transaction 5: {B, D, E} Calculate the support, confidence, and lift for the association rule {B} → {D}. Show all steps and calculations in your solution.	10	L_4	CO2
3.	A retail store wants to predict whether a customer will make a purchase based on their browsing behavior. The store collects the following data:	10	L_4	CO2

Time Spent on Site (minutes)	Number of Pages Visited	Made Purchase (Class Label)
5	3	Yes
2	1	No
8	5	Yes
1	1	No
7	4	Yes

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INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**28th January 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM73GC**	**Duration**	**120 Mins**
Sem	**VII Semester**	**CIE – II**	
VIRTUAL COMMERCE (Departmental Elective)			

Note: Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	What is meant by a protocol in an internet architecture context?	02	L2	CO1
2	What is internet economy?	02	L2	CO1
3	Mention any two features of B2G e-commerce model	02	L1	CO2
5	Uniform resource locator (URL) is used for providing _____ for a web page.	01	L2	CO2
6	Apart from considering factors relating to the firm's competitive environment, it is also necessary to consider factors relating to the firm's _____	01	L2	CO1
7	Social commerce is a concept where electronic commerce is used in conjunction with _____	01	L1	CO2
8	Any business that expects to grow profitably and effectively will have to master the art of selling through a _____ of multiple channels.	01	L1	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	"Among the more evident benefits of e-markets is the increase in price transparency." Justify this statement with suitable case example	10	L3	CO3
2.	Evaluate the three major forces fueling e-commerce?	10	L4	CO4
3.	How can mass customization concept impact the efforts of taking any business to V-commerce platform?	10	L2	CO2
4.	Explain the advantages of presence of V-commerce in multichannel marketing.	10	L2	CO1
5.	Explain the driving Factors of B2C Online-Selling applied to Indian markets, targeting rural customers	10	L2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	06	04	–	–	–	04	06	–	–	–	–
		Test	10	20	10	10	–	–	30	10	10	–	–

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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**29/01/2025**	**Maximum Marks**	**10 + 50**
Course Code	**21IM75IN**	**Duration**	**30 + 90 Min**
Sem	**VII**	**CIE – II**	
SUPPLY CHAIN ANALYTICS (Institutional Elective II)			

Note:
1. Answer all the Questions.
2. Use of Statistical table permitted

Part – A

Sl. No.	Questions	M	BT	CO
1	Enumerate any 4 supply chain risks.	02	02	03
2	Identify any 4 important factors to be considered for warehouse layout design.	02	02	03
3	A company operates with an average lead time of 4 days and a daily average demand of 100 units. The calculated safety stock is 476 units. Calculate the Reorder Point (ROP).	02	03	03
4	What are the four main stages of the SPSS model in demand management?	02	02	04
5	State the formula used for calculating MAPE in demand forecasting.	02	02	04

Part –B

Sl. No.	Questions	M	BT	CO
1	Choose a manufacturing firm and a service provider, and compare and contrast which one has a higher degree of vertical integration, and which one has more outsourcing? Discuss the pros and cons of vertical integration and outsourcing with your chosen companies	10	3	2
2	Managing relationships in a multi-tier supply network can be very complex and challenging. Discuss the various relationship types in a network and approaches to managing them.	10	2	3
3	Discuss the role of a Warehouse Management System (WMS) in optimizing warehouse operations. Highlight its key benefits with examples. Additionally, identify and explain five Key Performance Indicators (KPIs) commonly used to measure warehouse performance and their significance in improving operational efficiency.	10	3	3
4	Analyze the various demand forecasting methods by categorizing them into qualitative and quantitative approaches. Examine the key differences between these two types of methods and assess how they impact decision-making in different business scenarios.	10	4	4
5 a	A company operates with an average daily demand D_{avg} of 100 units and an average lead time LT_{avg} of 4 days. The standard deviation of daily demand σ_D is 20 units, and the standard deviation of lead time σ_{LT} is 2 days. The company aims to achieve a 95% service level. Using the given data, calculate the safety stock required to meet the target service level. Z value is 1.96	05	2	3
b.	Holding inventories can be costly. Identify the various costs associated with inventory holding within a company of your choice	05	3	3

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Academic year 2024-2025 (ODD Sem)

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	–	–	06	04	–	08	02	–	–	–
	Test		–	10	30	10	–	15	25	10	–	–

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Department of Electronics & Telecommunication Engineering

Date	29th Jan. 2025	Maximum Marks/Quiz	50+10
Course Code/Title	21ET75IK	Duration	120 Min
Sem	VII Semester	CIE-II	

SPACE TECHNOLOGY AND APPLICATIONS

Q.No	PART A	M	BT	CO
1.	Mention the spectrum range and application of Mid Infrared bands in Remote sensing?	1	L2	CO3
2.	What is the difference between supervised and unsupervised image classification.	1	L2	CO3
3.	The 16 bit image consists of—- –number of pixel values.	1	L1	CO3
4.	The function of Handoff is_____	1	L2	CO2
5.	Communication satellites and weather satellites are often placed in _____Orbit.	1	L2	CO2
6.	RADAR is —— type of remote sensing.	1	L2	CO3
7.	What is temporal and Radiometric resolution?	2	L2	CO3
8.	List any two recent Indian satellites that are designed for extra-terrestrial scientific observations.	2	L3	CO4
	PART B			
1.	What are LEO, MEO and GEO orbits? Explain with relevance to applications. Mention at least one example of a satellite launched by India to each of the three orbits.	10	L2	C2
2. a)	Draw the block diagram showing a home terminal for DBS TV/FM reception and explain its principles.	06	L2	C3
b)	The orbit for an earth orbiting satellite has an eccentricity of 0.15 and a semi major axis of 9000 km. Determine: i) The Apogee height. ii) The Perigee height. (Assume R=6370 km.)	04	L3	C2
3:	Describe the functioning of Attitude and Orbit Control (AOCS) system in satellite and also explain different methods are used to control.	10	L3	C2
4.a)	Differentiate four types of multiple access techniques used in satellite transmission.	05	L2	C2
b)	Explain the principles of VSAT and their applications.	05	L2	C2
5.	What is Remote sensing? Explain two types of Remote sensing instruments. List out important advantages and disadvantages of Remote sensing.	10	L2	C3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
[illegible] (rest of the table is cut off in the scan)											

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Department of Electronics & Communication Engineering

Date: 7/2024	**Test - 2**	**Max. Marks:10+50**
Semester:VI	**UG**	**Duration: $1\frac{1}{2}$ Hrs**
Course: Human Machine Interface(Institutional Elective)		**Code: 21IE6F10**

S No	Part A	M	BT	CO
1.	UX concentrates on————————-	1		
2.	Simple sketches or digital mockups that show the layout, structure, and organization of content on a webpage or app screen are called————————	1		
3.	Name any 2 prototyping tools for UX————————	1		
4.	Fictional characters that represent the target users of a product, used to guide design decisions and keep the user's needs in mind are known as————————	1		
5.	The style and formatting of text used in a product, including font families, sizes, and weights is called————————	1		
6.	Cross-platform application programming interface (API) for rendering 2D and 3D vector graphics is called————————	1		
7.	Which of the following principles is NOT a key component of a user-cantered HMI development process————————————-	1		
8.	In the context of a user-centered HMI development process, which of the following methods is primarily used to gather detailed requirements directly from users?	1		
9.	Which of the following best describes the principle of 'Feedback' in mobile UI design?	1		
10.	Which of the following is NOT a common function of a web server?	1		

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Date: .07.2024	**Test – II**	**Max*[illegible]***
Semester: VI	**UG**	**Dura*[illegible]***
Course: Human Machine Interface(Institutional Elective)		**Code*[illegible]***

S No	Questions
1.a	List and explain the 5 elements of UX in detail with neat diagrams.
2. a	Illustrate the five core components of UX designing in detail.
3.	Illustrate the seven phases of Design Thinking process in detail.
4. a	List and briefly discuss the following: i. 5 dimensions of interactive design ii. Any 5 UX design asset types
b	What is a web server? Describe in brief the main components of a web server
5.	List any three types of HMIs. List the main stages of the user-centered HMI development process and explain the importance of each stage

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Academic year 2024-2025 (Odd Sem)

Common to all branches

DEPARTMENT OF
BIOTECHNOLOGY

Date	**January, 2025**	**Maximum Marks**	**10 + 50**
Course Code	**21BT75IB**	**Duration**	**120 min**
Sem	**VII Semester**	**Closed Book Offline Quiz and Test – 2**	
Healthcare Analytics (Institutional Elective)			

Answer all questions in Part A & Part B, Answer Part A in first two pages in sequence only

Part A

Sl No	Questions	M	BT-L	CO
1	Multiple sequence analysis consists _____ or more number of sequences.	1	2	1
2	Needleman_Wunsch algorithm can be applied for _____ alignment.	1	2	1
3	Expand BLAST.	1	2	2
4	Among the PAM1 and PAM250, which one has the more distant?	1	3	2
5	Differentiate between the opening and extension gap penalties.	2	2	3
6	What is null hypothesis?	2	4	3
7	Mention any 2 measures of statistical significance.	2	2	4

Part B

Sl. No	Questions (Test)	M	BT-L	CO
1.	What are the different types of sequence alignments? Explain any one in brief.	10	2	3
2	How does the scoring matrix helps to derive the interpretation among sequence alignment?	10	3	3
3	Elucidate on the profiles and Hidden Morkov Models for bioinformatics data.	10	4	4
4	Differentiate between PAM and BLOSSUM and write its significance.	10	4	4
5	Write an overview of molecular phylogeny.	10	2	3

BTL-Blooms taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	*illegible* 2	2	~~60~~ 34	~~40~~ 22		~~40~~ 27	~~20~~ 18	~~40~~ 20		

(Struck-out printed values were corrected by hand on the original.)

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Department of Electrical & Electronics Engineering

Academic Year 2024-25 (ODD SEM)

Date	29th Dec 2024	Maximum Marks	50
Course Code	21EE75IH	Duration	90 Mins
Sem	VII Semester	TEST-II	
E-Mobility			

Q.No	Part B – Test Questions	Marks	COs	BT
1a.	Discuss the different techniques employed in BMS to measure the following battery parameters: (a) Current and (b) Temperature	06	02	01
1b.	Why battery management systems have to be designed for EV? Explain briefly.	04	01	01
2a.	With a neat circuit diagram explain the process of active cell balancing process.	06	02	03
2b.	Explain briefly about the causes of cell imbalance.	04	02	01
3a.	Analyze in detail about CV chargers.	06	03	03
3b.	What are balancers? Explain its applications.	04	02	04
4a.	Discuss any three issues related to energy management strategies implementation.	06	03	02
4b.	Discuss briefly about the classification of various energy management strategies.	04	01	01
5a.	Explain in detail about deep reinforced learning energy management algorithm.	06	03	03
5b.	Compare the following Energy Management strategies: (a) rule-based strategy and (b) Optimization Based.	04	03	03

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	08	20	22	-	18	06	22	04	-	-

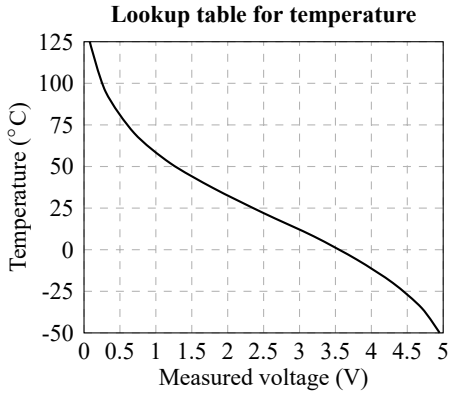
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Department of Electrical & Electronics Engineering

Academic Year 2024-25 (ODD SEM)

Date	29th Dec 2024	Maximum Marks	*[illegible]*
Course Code	21EE75IH	Duration	*[illegible]*
Sem	VII Semester	QUIZ-II	
E-Mobility			

Q.No.	Part A – Quiz Questions	Marks	COs	BT
1.	Define energy management strategy in the context of electric vehicles.	02	2	*[illegible]*
2.	List the essential functions of a BMS.	02	2	*[illegible]*
3.	What are the primary inputs for BMS to estimate the SOC.	02	2	*[illegible]*
4.	Why are energy management strategies critical in hybrid and electric vehicles?	02	3	*[illegible]*
5.	By considering the below graph identify which type of thermistor is employed for temperature measurement 	01	3	*[illegible]*
6.	What happens if the battery is allowed for overcharging?	01	3	*[illegible]*

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	06	04	-	06	03	01	-	*[illegible]*	*[illegible]*

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Department of Civil Engineering

Date	29-01-2025	Maximum Marks	50+10
Course Code	**21CV75IF**	Duration	120 Min
Sem	VII	CIE – II	
INTEGRATED HEALTH MONITORING OF STRUCTURES			

Note: Answer all questions.

QUIZ

Sl. No.	Questions	M	BT	CO
1	Define smart materials in the context of Structural Health Monitoring (SHM)	2	1	2
2	What is Static field testing	2	1	3
3	List two Electric Field Sensors	2	1	2
4	List the Benefits of AI in SHM	2	1	2
5	Load cell can be used as _____, _____	2	1	2

TEST

Sl. No.	Questions	M	BT	CO
1	With a neat sketch explain the evolution of materials used for structural health monitoring and its application .	10	3	2
2	Classify Smart Materials used in SHM with examples	10	3	2
3	Explain the different types of static load tests and conditions for its application	10	2	3
4	Discuss the sensor systems and hardware requirements in Static Field Testing	10	2	3
5	Explain the working Principle and advantage of EMI Technique for structural health monitoring	10	2	2

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	38	22	-	10	30	20			

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Academic year 2024-2025 (Odd Sem)

(CIE-II FOR VII SEM)

DEPARTMENT OF CHEMISTRY

Date	29.01.2025	Sem - VII	CIE-II
Course Code	21CM75ID	Maximum Test Marks	Quiz-10 + Test-50 (60M)
Course Name	ACSM	Duration	120 mins
ADVANCES IN CORROSION SCIENCE AND MANAGEMENT			

Note: 1) Answer all the questions

2) Part A should be answered in the first two pages of the answer script.

	Part-A	**M**	**BTL**	**CO**
1	What is the impact of substrate temperature on the PVD (sputtering) process?	1	4	2
2	How does the alkalinity of an environment influence the interpretation of Pourbaix diagrams?	1	3	2
3	What is the role of resin in varnish?	1	2	1
4	How do you differentiate between varnish and enamel?	1	3	3
5	Chromium anodes are not used in the process of chrome plating. Justify	1	3	2
6	What are oxygen scavengers?	1	2	1
7	Identify the missing product in the following reaction concerning the electroless plating of Nickel. $H_2PO_2^- + H_2O + Ni^{+2} \implies ? + 2H^+ + Ni$	1	4	2
8	Write the electrolytic bath composition of Nickel plating.	1	1	2
9	Name the film-forming material in paint.	1	1	2
10	List any one limitation of the Pourbaix diagram.	1	1	3

	Part-B	**M**	**BTL**	**CO**
11	Outline the methods by which corrosion inhibitors function to reduce corrosion rates in metal structures. Explain with reactions.	7	2	2
12	For the oxygen reduction reaction $O_2(g) + 4\ H^+(aq) + 4e^- \longleftrightarrow 2H_2O\ (l)$ The standard electrode potential (Eo) is +1.23 V. Calculate the equilibrium potential at pH= 7, where the concentration of oxygen is 0.21 atm. Interpret the result obtained.	7	3	3
13	What are the key differences between electroless plating and electroplating? Explain the process of copper electroplating with reactions.	7	*[illegible]*	*[illegible]*
14	List out the main components of paints. Explain any 3 components with examples along with their role.	7	*[illegible]*	*[illegible]*
15	How can corrosion be controlled through effective design? Analyze and interpret the role of design in corrosion control concerning practical situations. what is the clarity of this question	7	*[illegible]*	*[illegible]*

16	An experiment involved two aluminum samples, each with an initial weight of 200 grams. One sample was treated with a corrosion inhibitor, while the other remained untreated. After exposure to a marine environment for 30 days, the treated sample's final weight was 198 grams, and the untreated sample's final weight was 195 grams. Given the density of aluminum is 2.7 g/cm^3 and the constant K is 87.6, calculate the corrosion rate for each sample using the weight loss method, expressed in millimeters per year (mmpy). Compare and interpret the results to determine the effectiveness of the corrosion inhibitor.	7	3	*[illegible]*
17	Construct a Pourbaix diagram for Aluminium, assuming the activity of all dissolved substances is 10^{-6} M at room temperature (298K). Using the diagram, interpret the corrosion tendency of Aluminium under different conditions, including the relevant electrochemical reactions. Discuss practical solutions for mitigating Aluminium corrosion based on your diagram analysis.	8	3	*[illegible]*

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

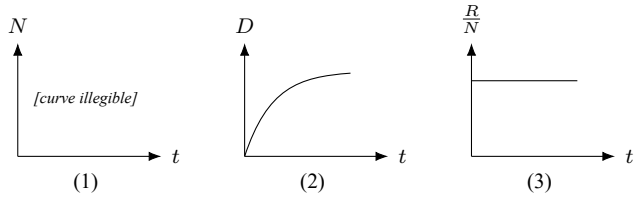
Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	*[illegible]* (cut off in the scan)
	Max Marks	16	13	16	15	4	16	25	15	

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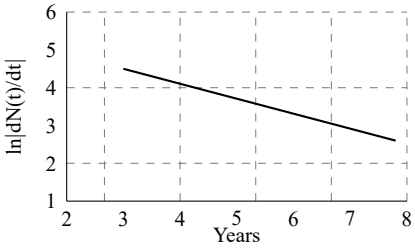
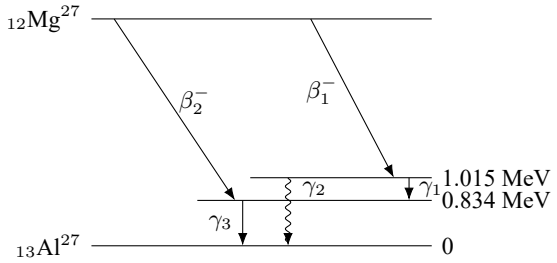
Department of Mechanical Engineering

Date	**29 January 2025**	Maximum Marks	Quiz 10 + Test 50
Course Code	**21ME75IQ**	Duration	120 Min
Course Name	**Nuclear Engineering**	**USN:**	

Q	PART-A (Quiz)	M
1.1	Calculate the binding energy of an alpha particle if mass of $^{1}_{1}$H atom is 1.007826 u, mass of neutron is 1.008665 u, mass of $^{4}_{2}$He atom is 4.00260 u. Take 1 u = 931 MeVc^{-2} .	1
1.2	Find the increase in mass of water when 1 kg of water absorbs 4.2×10^3 J of energy to produce a temperature rise of 1 K.	1
1.3	The binding energy of $^{35}_{17}$Cl nucleus is 298 MeV. Calculate its approximate atomic mass. Given that, mass of proton is m_p = 1.007825 amu and mass of neutron is m_n = 1.008665 amu.	1
1.4	The electric potential energy (in **MeV**) of interaction due to the electric repulsion between two nuclei of ^{12}C when they touch each other at the surface is given by…… Assume that the potential energy of interaction between two nuclei is given by U = $\frac{1}{4\pi\epsilon_0}\frac{Q_1Q_2}{r}$.	1
1.5	A radioactive nucleus X decays to a stable nucleus Y. Then the graph of rate of formation of Y against time t will be given as	1
1.6	A nucleus X initially at rest, undergoes alpha decay according to the equation $^{232}_{Z}$X $\rightarrow$ $^{A}_{90}$Y + α. What fraction of the total energy released in the decay will be the kinetic energy of the alpha particle?	1
1.7	At time t = 0, N_1 nuclei of decay constant λ_1 and N_2 nuclei of decay constant λ_2 is mixed. The decay rate of mixture is	1
1.8	Sun radiates energy in all direction. The average energy received at earth is 1.4 kWm^{-2}. The average distance between the earth and the sun is 1.5×10^{11} m. If this energy is released by conservation of mass into energy, then the mass lost per day by the sun is approximately	1
1.9	Nuclei of radioactive element A are produced at a rate t^2 at any time t. The element A has decay constant λ . Let N be the number of nuclei of element A at any time t. At time t = t_0, dN/dt is minimum. Then the number of nuclei of element A at time t = t_0 is	1
1.10	In a radioactive decay, let *N* represent the number of residual active nuclei, *D* the number of daughter nuclei, and *R* the rate of decay at any time *t*. Three curves are shown in figure. The correct ones are  (1) (2) (3)	1

Department of Mechanical Engineering

Q	PART-B (Test)	M
1	In an nuclear lab an experiment is conducted and following result is observed: A radionuclide with half life 1620 sec is produced at a constant rate 1000 nuclei per second. During each decay energy 200 MeV is released. If production of radio nuclides started at t = 0, calculate (i) the rate of release of energy at 3240 second and (ii) the total energy released upto 405 second.	10

2	During an EL presentation on nuclear engineering topic at RVCE to determine the half-life of a radioactive nuclide a group of students shown the graph of $\ln\left\|\frac{dN(t)}{dt}\right\|$ versus t as shown below. It was mentioned that $\ln\left\|\frac{dN(t)}{dt}\right\|$ represents the rate of radioactive decay at time t. If the number of radioactive nuclei of this element decreases by a factor of *p* after 4.16 yr , then what was the calculated value of *p*?  	10
3	For a radioactive material, its activity *A* and rate of change of its activity R are defined as A $= -\frac{dN}{dt}$ and R$= -\frac{dA}{dt}$, where $N(t)$ is the number of nuclei at time t . Two radioactive source *P* (mean life, τ) and *Q* (mean life, 2τ) have the same activity at $t = 0$. Their rate of change of activities at $t = 2\tau$ are R_P and R_Q , respectively. If $\frac{R_P}{R_Q} = \frac{z}{e}$, calculate the value of z.	10
4	The β-particles and γ-decays happens as shown in the figure below:  Find the kinetic energy of β-particles and radiation frequencies of γ-decay. Given that mass of $^{27}_{12}Mg$ atom is 26.991425 u and mass of $^{27}_{13}Al$ atom is 26.990080 u.	10
5	The nucleus X is unstable was initially at rest and after decay of an α-particle becomes stable Y nucleus as shown: $^{228}_{92}\text{X} \rightarrow {}^{A}_{90}\text{Y} + \alpha$. At $t = 0$, the emitted particle enters in a region of space where a uniform magnetic field $\vec{B} = \text{B}_o\hat{i}$ and electric field $\vec{E} = \text{E}_o\hat{i}$ exist. The α-particle enters in the region with velocity $\vec{V} = \text{V}_o\hat{j}$ from x = 0. At time t = $(\sqrt{3} \times 10^7)\left\{\frac{m_\alpha}{q_\alpha E_o}\right\}$ seconds, the particle was observed to have speed twice the initial speed v_0 , then find (a) the velocity of a particle at time *t*. (b) the initial velocity v_0 of the α-particle, (c) the binding energy per nucleon of α-particle. Given that: $m(Y) = 221.03$ u, $m(\alpha) = 4.003$ u, $m(n) = 1.009$ u, $m(p) = 1.008$ u, $m_\alpha = (2/3)\times 10^{-26}$ kg, $q_\alpha = 3.2 \times 10^{-19}$ C and 1 u = 931 MeVc^{-2}	10

All the best!

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Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF
CHEMICAL ENGINEERING

Date	**29th Jan 2025**	**Maximum Marks**	**60**
Course Code	**21CH751C**	**Duration**	**120 Min**
Sem	**VII Semester**	**CIE 2**	
SUSTAINABILITY AND LIFE CYCLE ANALYSIS (Institutional Elective)			

Note: 1. Answer all the Questions.

		Part A	M	BT	CO
1	a	Name the frame works used for the identification and structuring of environment indicators.	1	1	2
	b	Name the technique that can detect most of the elements of periodic table.	1	1	2
	c	Identify the component of a hair dryer that has the highest environmental impact.	1	3	3
	d	Define traditional engineering.	1	1	3
	e	List two contradictions of LCA.	1	1	2
	f	State Sherwood relationship.	1	1	2
	g	Name one each government data base and commercial data base available for LCA	2	1	1
	h	Analyze and calculate the ratio of the abundance of materials in the Earth's crust (tons) to the annual use of the materials (tons/yr). Assume an approximate for the crust of 2 x 10^{19} metric tons. This mass is based on a 40 km thickness for continental crust and a 3 km thickness for oceanic crust, with an average density of 3 g/cm^3 (3 tons/m^3). The availability of P in abundance is 0.1% and its usage of Tons/Year is 1.53 x10^6.	2	4	2
		Part B			
		List the components of LCA and explain the LCA process in detail.	10	2	3
2	a	With a neat diagram explain Inductively coupled plasma atomic emission spectroscopy ICP AES	07	2	4
	b	List the steps of toxicity characteristic leaching procedure.	03	1	2
3	a	With a neat diagram explain the sustainable development and design thinking.	07	2	3
	b	Explain the three-pronged approach to characterize the foot print	03	2	2
4		Construct an explanation of the chemistry of dry gasification.	10	3	3
5		Analyze the parameters of sustainability and explain why stone, leather, metals, and timber are regarded as sustainable materials.	10	4	4

Academic year 2024-2025 (Odd Sem)

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

Date	**Jan 2025**	**Maximum Marks**	**10+50**
Course Code	**21CS75IE**	**Duration**	**120 Min**
Sem	**07**	**CIE-II**	
PROMPT ENGINEERING (INSTITUTIONAL ELECTIVES – II (Group G))			

Q. No.	Questions	M	BT	CO
	QUIZ			
	1.1 Give one example where few-shot learning can be used in LLM applications	01	1	1
	1.2 Name one external tool commonly used with LLMs for real-time data retrieval.	01	2	2
	1.3 Which API would be useful for integrating weather data into an LLM application	01	2	2
	1.4 What is the benefit of using an external database for LLM-based question-answering tasks ?	01	1	1
	1.5 What type of external data would improve the performance of an LLM used for financial predictions ?	01	1	1
	1.6 Which factor is crucial for the successful deployment of prompts in production environments ?	01	1	1
	1.7 Give one example of a task where information extraction from social media could be useful?	01	1	2
	1.8 What does “NER” stand for in the context of information extraction?	01	2	1
	1.9 Provide an example of an entity that could be extracted from the sentence “Schedule a meeting with John tomorrow at 3 PM	01	1	1
	1.10 In the context of API integration, what does REST stand for ?	01	1	1
	TEST			
2.	Summarization using sources is an essential task for LLMs. How can LLMs leverage external sources to generate more precise and reliable summaries? Discuss the methodology for retrieving relevant information from external data sources, synthesizing it, and generating a coherent summary. Provide examples from industri*[illegible]*	10	1	1
3.	Explain the best practices in prompt engineering for large language models (LLMs). Discuss how prompt design influences model performance and how to craft clear, specific, and concise prompts for effective results. Provide examples of both effective and ineffective prompts.	10	1	2
4.	Advanced prompting techniques often involve the combination of multiple approaches. Discuss how advanced prompting methods like few-shot learning, zero-shot learning, and chain-of-thought prompting are applied in real-world scenarios. Compare their effectiveness and challenges in different use cases. task without any explicit examples, relying solely on the clarity of the instructions.	10	4	5
5.	Explain the role of information extraction in function calling. How can a conversational agent extract specific information (such as dates, times, and names) from user input, and how does it invoke functions to act on this extracted data?	10	3	4

Academic year 2024-2025 (Odd Sem)

6.	Explain the different techniques used in document parsing for information extraction. How can complex documents such as contracts, financial statements, and academic papers be parsed to extract meaningful information for automation and decision-making processes?	10	2	3

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BT-Blooms Taxonomy, CO-Course Outcomes

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test & Quiz	Max Marks	17	13	10	10	10	27	13	10	-	-

Course Outcomes:

CO1:	Demonstrate an understanding of prompt engineering principles including how prompt structure and phrasing impact the performance of AI models.
CO2:	Design and implement effective prompts- to create and apply prompts for various natural language processing (NLP) tasks, such as text generation, summarization, and translation, using AI models..
CO3:	Critically evaluate the effectiveness of prompts - assess the quality and performance of prompts in terms of accuracy, coherence, and relevance, identifying *[illegible]*
CO4:	Apply prompt engineering *[illegible]* (rest of the page is cut off in the scan)

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**29/01/2025**	**Maximum Marks**	**10+50**
Course Code	**21HS75IR**	**Duration**	**120 Mins**
Sem	**VII**	**CIE – II**	
PRINCIPLES & PRACTICES OF CYBER LAW			

Note:

1. Answer all the Questions.

Q.No.	**PART- A**	M	BT	CO
1	List any two types of E-commerce models.	2	2	CO2
2	What are 'Digital Signature' and its use in E-commerce?	2	2	CO2
3	What is the main concern regarding copyright infringement in cyberspace?	2	2	CO3
4	How does cybersquatting affect trademark owners?	2	2	CO3
5	How can a trademark owner protect their brand online?	2	2	CO4
	PART -B			
1	A company named XYZ Ltd. enters into a contract with ABC Ltd. through an online platform. The contract is electronically signed by representatives of both parties using Digital Signatures issued by a recognized Certifying Authority. However, later, XYZ Ltd. disputes the validity of the contract, claiming that one of the signatories had not authorized the use of their digital signature. (i)What role does a Digital Signature play in validating an electronic contract? (ii)What legal provisions under the IT Act 2000 support the enforceability of this electronic contract?	10	4	CO2
2	A company launches a new product and discovers a similar domain name being used to divert customers to a competitor's site. Propose a comprehensive strategy to address this issue legally and technologically with one case example.	10	4	CO2
3	Bring out the difference between Digital Signature & Electronic Signature along with the advantages and disadvantages.	10	2	CO3
4	Design a comprehensive strategy for an e-commerce business to protect customer transactions using digital signatures and cryptography.	10	3	CO4
5a	If an individual finds that their copyrighted work has been uploaded to a website without their consent, what steps should they take to protect their rights.	05	2	CO4
5b	Analyze the impact of piracy on digital content creators in cyberspace.	05	2	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	30	30				20	10	20		

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**28.01.2025**	**Maximum Marks**	**05 + 25**
Course Code	**21IM78**	**Duration**	**60 Mins**
Sem	**VII Semester**	**CIE – II**	
INNOVATION AND ENTREPRENEURSHIP			

Note:

1. Answer all the Questions.
2. Use suitable sketches and tabular column wherever necessary.
3. Observe the marks split-up and answer accordingly.

PART – A

Sl. No.	Questions	M	BT	CO
1	List any two rules of brainstorming.	01	L1	CO1
2	Imagine you are starting your ice-cream outlet. List 2 ways in which you can make sure your funds are used optimally.	01	L2	CO2
3	Identify two bad practices students are doing to get "AICTE activity points". Suggest two methods to curtail the same.	02	L4	CO2
4	Comment on the importance of "Early Adopters".	01	L3	CO3

PART – B

Sl. No.	Questions	M	BT	CO
1.	Discuss the four types of Markets (Use tabular column format) and comment on the most suitable market type for a start-up venture.	05	L2	CO2
2.	When you are allowed to form teams on your own, lack of coordination among the team members is clearly visible and reflects in the outcome of the work also. As individuals, each of the students is competent in their own way. Use design thinking principles to identify the problem with students and how it can be resolved. (Concentration should be on the first three stages of DT)	05	L4	CO3
3.	Illustrate the lean canvas of business model for your practice venture using Ash Maurya's template and assess each of the 9 blocks using a 5-point scale (1 being lowest and 5 being highest). Identify any 3 blocks which needs improvements and suggest strategies to improve the same.	05	L3	CO4
4.	Construct 2 detailed personas of students belonging engineering education from your college and propose 2 problem definitions using DT framework.	07	L5	CO3
5.	Choose any 1 problem definition from the previous question and answer the following. i. Assess & depict the input requirements based on your understanding the solution space. (2 marks) ii. Sketch the solution. (1 mark)	03	L5	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
Marks	Max Marks	Quiz	01	03	01	–	–	01	01	01	02	–	–
		Test	–	05	12	08	–	–	05	05	05	10	–

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